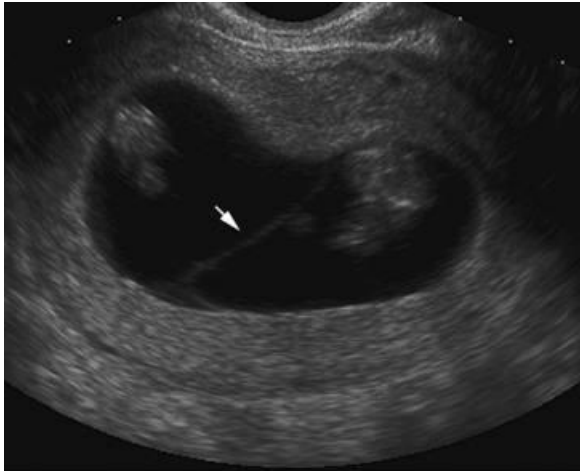


1. **D**
 2. **C** - Regardless of what her symptoms are or what the actual diagnosis is, pregnancy **MUST** be ruled out and therefore C is the correct answer. **Q:** Shouldn't it be urine hCG? **A:** Either can be used to diagnose pregnancy. Since urine isn't an option pick the one that is.
 3. **B** - Previous macrosomic infant is risk for development of gestation DM. In these patients we check glucose at first antenatal visit.
 4. **B** - Confirm syphilis infection with either screening test when one test shows positive result.
 5. **B** - One of the main risk factor for endometrial carcinoma is **OBESITY** especially in postmenopausal. This is mainly because of the peripheral conversion of androgens into estrogens and aromatization of androstenedione into estrone. This will lead to chronic stimulation of the endometrium by the estrogen without the balancing effect of progesterone (just like the effect of anovulation on the endometrium).
 6. **C** - Pelvic organ prolapse centrally = enterocele.
 7. **C** - Rh negative with antibody also negative that means C is the answer, daycare is clue to Parvovirus B19.
 8. **A** - Cervical cancer screening recommended twice in the first year after diagnosis of HIV and then annually if results are normal (UTD).
 9. **A** - Fetal bradycardia - pelvic exam to palpate cord prolapse.
 10. **B** - She can use Lamotrigine for seizures.
 11. **B** - Müllerian duct is responsible for the formation of 1. Uterus, 2. Fallopian tubes, and 3. Upper portion of the vagina. In müllerian duct agenesis the ovaries are completely fine and present, therefore a person with müllerian duct agenesis should present as amenorrhea with fully developed secondary sexual characteristics (indicator of functional ovaries). But in this patient, her secondary sexual characteristics are not fully developed (breast) therefore this patient is most likely an XY with decreased testosterone sensitivity which has led to the prominence of the estrogen and the formation of a rudimentary vagina (they usually have testes as well, but the testes should be removed to avoid malignancy). **Q:** But she has no uterus; why this is not D? **A:** In müllerian agenesis, secondary sexual characteristics (axillary or pubic hair) are normal.
 12. **B** - There are several things to look for that are dichorionic in nature or at least lend to it more like being dichorionic: >2mm dividing membrane, two genders, two placentas, twin peak (lambda sign). This one has 2 so it's very likely dichorionic.
- More:** So apparently, thanks to Google, I found out that the thickness is the only way to differentiate monochorionic from dichorionic. The first trimester is the best time to evaluate the chorionicity and amnionicity of twin gestations. Look at the difference in the thickness in these two images:
- A. Sonogram in a 10-week monochorionic-diamniotic gestation. A thin dividing membrane (arrow) from the two apposed amniotic sacs is seen.
 - B. Sonogram from a 7-week dichorionic-diamniotic gestation. Both amnions (arrows) surrounding the developing embryos are well seen in this diamniotic twin pregnancy. Dichorionicity is confirmed by the thick intertwin membrane.



A.



B.

13. **E**

14. **E** - Endometriosis, laparoscopy needed to confirm the diagnosis.

15. **D** - Patients with a positive TST or IGRA must undergo clinical evaluation to rule out active tuberculosis; this includes evaluation for symptoms (e.g. fever, cough, weight loss) and radiographic examination of the chest (with appropriate shielding), regardless of gestational age (UTD).

16. **J** - Amniotic fluid embolism syndrome happens either during the labor and delivery or in the immediate postpartum. In UTD they say that it rarely has been reported as late as 48 hours after a C-section. For this patient it has been 72 hours. Another clue is that amniotic fluid embolism presents with DIC. **Q**: Her fibrin split product is normal - it rules out PE. It has to be amniotic fluid embolism? **A**: Fibrin split products not as relevant as D-dimer for PE. **Q**: What about bilateral wheezing? Lungs should be clear? **A**: Her bilateral wheezing is probably due to the cor pulmonale. **Note**: no explanation why she is acidotic. She should have a respiratory alkalosis. Possibly a mistake.

17. **E** - Her chronic HTN leads to less than optimal placental blood vessels leading to uteroplacental insufficiency which leads to fetal growth restriction.

18. **C** - One important clue here is the "prolonged" labor in the very first line, which places her at greatest risk for endometritis. Diffuse cramping of lower abdomen + tenderness and rigidity on examination + fever + leukocytosis - all these are very typical for the diagnosis of postpartum endometritis. RBCs in the urine in the presence urinary catheter is suggestive of traumatic catheterization finding upon insertion (not significant to the given stem just a DISTRACTOR). RBCs in the urine is often a distractor in surgical questions.

19. **F** - She is a smoker and >35, both of which are contraindications for OCPs.

20. **A** - Hypogonadotropic hypogonadism due to extreme exercise.

21. **E** - Most common cause of abnormal uterine bleeding in young children. Runny nose is a distractor.

22. **A** - Urinary urgency + nocturia = urge incontinence (mainly secondary to detrusor instability). Detrusor instability may be caused by stones, infection, or in this case, a mass, where the bladder lining is irritated and produces a contraction. Incontinence in this patient is an indirect complication of the fibroid, not direct. 80% of women will have fibroids by 60, so when you see a 3 cm fibroid it shouldn't be the first thing you suspect. If it had been 13 cm then yes, but 3 cm seen on US tells you nothing because it's small and US cannot differentiate between fibroid location.

23. **D**

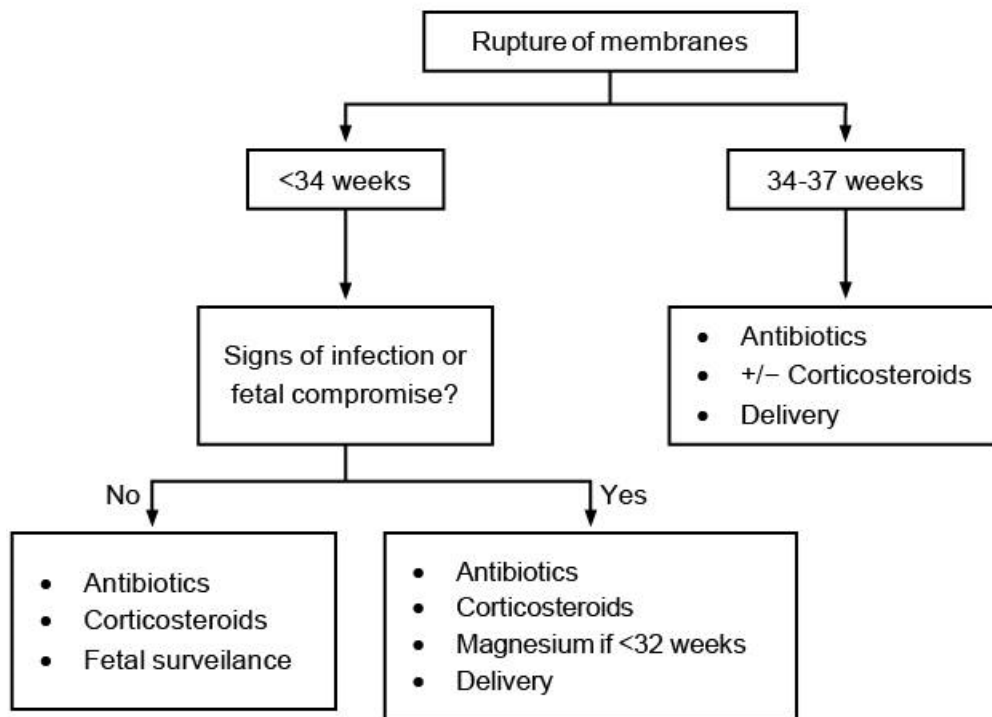
24. **A** - Here the best option is to administer the second dose of Betamethasone (which should be given in two doses within 24 hours)

In case of this option has not been listed, CHECK the gestational age (whether less than 34 weeks or more). This patient in this given scenario is 28 week gestation, presenting with preterm premature rupture of membrane (PPROM). As per UWorld recommendation, we should evaluate first for signs of infections or fetal compromise.

If these were POSITIVE: start with antibiotics + corticosteroids + MgSO₄ (if less than 32 weeks) + delivery

If signs were NEGATIVE: start with antibiotics + corticosteroids + fetal surveillance.

Management of PPRM



PPROM = preterm premature rupture of membranes.

25. **D** - Always think of intraductal papilloma when there is spontaneous discharge. Duct ectasia has a small mass behind areola and streaks are seen on imaging.
26. **C** - 5-alpha reductase deficiency. No conversion of testosterone to dihydrotestosterone, which is in charge of development of external male genitalia. They are raised as girls but in puberty due to increase in testosterone they acquire male characteristics. **More:** 5-alpha reductase converts testosterone to dihydrotestosterone. The latter is the most potent form, which is in charge of development of external male genitalia, male baldness, and enlargement of the prostate. Deficiency of this enzyme prenatally results in hypospadias and ambiguous genitalia.
27. **D** - SCC of cervix associated with benign retroperitoneal fibrosis and hydroureter (terrible triad of advanced cervical cancer is sciatic back pain, leg swelling, and hydroureter).
28. **C** - HPV 6 and 11: genital warts. Exposure to lower grade HPV strains likely coincided with exposure to the higher grade strains that gave her the LSIL. **More:** Genital warts commonly manifest as flesh-colored, painless, cauliflower-like lesions (condyloma acuminata), but may also manifest as smooth, dome-shaped papules (papular warts), flat crusty papules resembling seborrheic keratosis (keratotic warts), or macular or slightly raised flat-topped papules.
29. **A** - Give GBS prophylaxis if unknown status, premature, or delivery >18 hrs after RoM.
30. **F** - Acanthosis nigricans ("velvety pigmented skin") is associated with insulin resistance.
31. **C** - Maternal fever causes fetal tachycardia.
32. **D** - No fluid seen on US. **Note:** Explanation unsatisfactory.
33. **A** - Asherman Syndrome from the D&C.
34. **E** - In vulvar squamous cell carcinoma, most patients present with a unifocal vulvar plaque, ulcer, or mass (fleshy, nodular, or warty) on the labia majora; the labia minora, perineum, clitoris, and mons are less frequently involved. In lichen sclerosis the patient presents with a whitish plaque.
35. **C** - Cervical stenosis and/or insufficiency from LEEP x2.
36. **C** - Endometrial hyperplasia, treat with cycle progestin therapy to shed the endometrium.
37. **D** - Diabetics are at risk for polyhydramnios. **Q:** Even when the blood sugar is control? **A:** HbA1c would reflect her glycemic state for longer period. **Q:** How to rule out error in gestational age? **A:** Because 1st trimester ultrasound estimates gestational age accurately and she started receiving prenatal care since 7 weeks so dating error is unlikely.
38. **A** - Smoking is a risk factor for IUGR and due to decreased fundal height, do a NST. A fetal fibronectin is done at <34 weeks of gestation together with cervical length measurement.
39. **A** - CAH leading to virilization, not AIS because AIS is XY and option D is a normal process of X inactivation in females.
40. **C** - Chancroid due to *H. ducreyi*.
41. **D**

Screening men: The U.S. Preventive Services Task Force (USPSTF) strongly recommends screening men 35 years and older for lipid disorders (A), men 20 to 35 years of age for lipid disorders if they are at increased risk of coronary heart disease (CHD) (B).

Screening women at increased risk: The USPSTF strongly recommends screening women 45 years and older for lipid disorders if they are at increased risk of CHD (A), women 20 to 45 years of age for lipid disorders if they are at increased risk of CHD (B).

Screening young men and all women not at increased risk: The USPSTF makes no recommendation for or against routine screening for lipid disorders in men 20 to 35 years of age, or in women 20 years and older who are not at increased risk of CHD (C).

42. **G** - She is sexually active with no other screening indications.

43. **A** - Patient had miscarriage but is stable, thus hCG needs to be followed until it is undetectable.

44. **F** - Pulmonary hypoplasia due to early gestational age and polyhydramnios.

45. **A** - Severe presentation of (fever, nausea) is among indications for hospitalization for PID.

46. **D** - She likely has trichomoniasis. Absences from school are supposed to be a clue to sexual activity. **More:** Ectropion occurs when eversion of the endocervix exposes columnar epithelium to the vaginal milieu. It is common in adolescents. After adolescence, it may be observed in women who are pregnant or taking estrogen-progestin contraceptives or who had a cervical laceration during labor and delivery.

47. **B** - 1.) Hysterosalpingogram is far less invasive than surgery; 2.) We know she's ovulating because she has regular cycles, no ovulation = CL = no progesterone = no progesterone withdrawal = no period. Therefore we know she has estrogen, we know she's ovulating, we know the sperm is good to go, now we just need to know if there is an anatomic abnormality and given a history of multiple partners the best test to evaluate = HSP.

48. **B** - Hypochromic (low MCH), microcytic anemia, normal HbA % = IDA.

49. **D** - Ultrasound is needed, but what's interesting is gestational age is for sure accurate as it is not going off of LMP since she has already experienced menopause. She was implanted with a viable fertilized egg at a known date, so ultrasound is used to evaluate why the fundal height is not matching up with her gestational age (e.g. multiple gestations).

50. **J** - A differential diagnosis for postpartum fever includes pulmonary or amniotic fluid embolism (dyspnea, tachycardia, V/Q mismatch), UTI/pyelonephritis (abnormal urinalysis, flank pain, CVA tenderness), endometritis (tender uterus), wound infection (wound indurated and erythematous), septic pelvic thrombophlebitis (diagnosis of exclusion – all other causes eliminated and unresponsive to antibiotics). Nothing fits her symptoms here besides a wound infection. She definitely has some sort of infection based on her fever and CBC. If her wound were “clean and dry”, then you would default to septic pelvic thrombophlebitis, which requires antibiotics and anticoagulants.

In next pages, you will find the answers made by someone else but the order of answers is not same. I have left it there in case it helps you better understand the answers.

NBME OBGYN Form 2 - Answers & Explanations

Updated: Feb 25, 2020

1) Enterocele

- Enterocele = descent of small intestines into lower pelvic cavity → pushes on top part of vagina (ie, origin is high in vaginal vault) → bulge
- Hysterectomy → ↑ risk of enterocele (removal of uterus → space available for intestines to descend)
- Rectocele → bowel movement problems (not seen in enterocele since it is the small intestines that descends)
- Cystocele → urinary tract sx. Cystocele would involve the anterior vaginal wall, not posterior wall

2) Fetal parvovirus B19 infection

- Patient works in a daycare (contact with children); Parvovirus B19 is most common infection associated with hydrops
- CMV and Toxo are less common causes of hydrops
- 'Antibody screening is negative' = Rh sensitization has not occurred

3) Iron deficiency

- ↓ MCV (microcytic)
- Physiologic dilution of pregnancy and sickle cell would both be normocytic
- Folic acid or B12 deficiency would be macrocytic
- Hb electrophoresis excludes sickle cell

4) Vulvar carcinoma

- Clinical features: vulvar pruritis, vulvar plaque/ulcer, abnormal bleeding
- Patients often have a unifocal friable mass, commonly located on the labia majora

5) Cesarean delivery

- Leopold's Maneuvers help determine position of fetus
- If transverse lie is recognized during active labor → C-section
- Can't perform external cephalic version (ECV) or internal podalic version if in active labor

6) Condyloma acuminata

- HPV 6, 11 → Condylomata acuminata
- Tx: Podophyllin resin, Trichloroacetic acid

7) Admission to the hospital for intravenous antibiotic therapy

- This patient has pelvic inflammatory disease (PID)
- Indications for hospitalization include high fever, inability to take oral meds, and risk of non-adherence to treatment

8) Pelvic examination

- If the baby is not in the correct position (ie, fetal station of at least 0) and there is sudden rupture of membrane (SROM), there is risk of cord prolapse (there is space for it to fall out; if baby is in correct position, the head would prevent cord prolapse)
- Cord prolapse presents with rapid decrease in fetal HR
- Pelvic examination will allow you to feel/see the cord to confirm your dx
- Tx: Immediate cesarean section

9) Intraductal papilloma

- MCC of unilateral bloody nipple discharge
- Serosanguinous = contains both blood and serum

10) Measurement of serum β -hCG concentration

- First step in work-up of secondary amenorrhea is to measure β -hCG to rule out pregnancy

11) Asherman syndrome

- $\ominus$ Progesterin challenge can be either estrogen deficiency (ovarian failure) or asherman
- This patient has D&C history and normal FSH which points to asherman syndrome

12) Second administration of corticosteroids in 24 hours

- This patient has preterm premature rupture of membranes (PPROM) - ruptured of membranes <37 weeks gestation (preterm) and before labor has begun (premature)
- A course of antenatal corticosteroids consists of betamethasone (~12 mg) intramuscularly every 24 hours for two doses
- MgSO₄ should also be give to prevent cerebral palsy

13) Maternal fever

- Fetal tachycardia is due to maternal fever (mother should be given antipyretic to reduce fever and subsequently improve the fetal tachycardia)
- Fetal anemia → sinusoidal pattern
- Fetal head compression → early decelerations
- Umbilical cord compression → variable decelerations
- Uteroplacental insufficiency → late decelerations

14) Squamous cell carcinoma of the cervix

- Advanced cervical carcinoma often invades through the anterior uterine wall into the bladder, blocking the ureters → obstructive uropathy (hydroureter)

15) Intravenous administration of ampicillin and gentamicin

- This patient has endometritis, which is usually treated with clindamycin + gentamicin due to its **broad-spectrum coverage**
- Ampicillin can be used instead of clindamycin if there is clindamycin resistance, $\oplus$ GBS, or if infection persists after starting antibiotics

16) Wet mount preparation of the vaginal fluid

- This patient has trichomoniasis
- Wet mount would show motile trichomonads

17) X-ray of the chest

- HIV patients with >5 mm induration = $\oplus$ PPD
- CXR is the next best step for $\oplus$ PPD patients with a risk factor (ie, HIV), regardless of gestational age

18) Third measurement of serum β -hCG concentration in 1 week

- This patient had an abortion and her β -hCG should be monitored weekly until it is undetectable

19) Pap smear

- Women with HIV should have a Pap test at the time of HIV diagnosis (baseline), then every 12 months (some recommend a second test at 6 months as well). If the results of 3 consecutive test are normal, follow-up Pap test should be every 3 years.

20) Triphasic oral contraceptives

- OCPs are contraindicated in smokers >35 years of age

21) ACTH oversecretion

- This patient has congenital adrenal hyperplasia $\rightarrow$ virilization

22) 46, XY

- 5-alpha-reductase deficiency
 - Male internal genitalia, Female external genitalia
 - Phenotypically female at birth with virilization at puberty

23) Uterine inversion

- Uterine fundus prolapses through the cervix/vagina $\rightarrow$ smooth round mass protruding through the cervix/vagina
- Uterine fundus will be non palpable transabdominally
- Tx: aggressive fluid replacement, manual replacement of the uterus, uterotonic drugs after uterine replacement

24) Dichorionic

- Thick septum = dichorionic (chorion = placenta)
- Two yolk sacs = diamniotic (amnion = amniotic sac/yolk sac)

25) Measurement of serum cholesterol concentration

- Father had MI at young age (39 yo) and she has other CVS risk factors (eg, obesity, smoking)

26) Testing for *Chlamydia trachomatis*

- All sexually active females < 25 years old should be offered annual *Chlamydia* and *Gonorrhea* screening
- This patient also uses condoms inconsistently

27) Pulmonary embolism

- This patient is 3 days postpartum, and thus PE is more likely (Virchow triad: hypercoagulable state from pregnancy, stasis from bed rest, and vascular insult)
- Amniotic fluid embolism would essentially occur in the immediate postpartum period
- Massive PE $\rightarrow$ circulatory collapse $\rightarrow$ metabolic acidosis

28) Torsion of an ovarian cyst

- Sudden abdominal pain in the context of an ovarian mass is torsion until proven otherwise
- Cystic and solid components = dermoid cyst, which has an $\uparrow$ risk of torsion due to heterogeneous composition

29) Detrusor instability

- This patient has urge incontinence with nocturia
- Fibroids would cause stress incontinence due to increased abdominal pressure $\rightarrow$ loss of urine with coughing, sneezing, valsalva
- Additionally, leiomyomas regress in the setting of menopause - if she had a fibroid, it was probably present for several years and shouldn't be causing a new onset of urinary sx's

30) Neural tube defects

- Anti-epileptic drugs (eg, Valproate, Carbamazepine, Phenytoin, Phenobarbital) $\rightarrow$ NTDs

31) Cervical incompetence

- Previous LEEP surgeries + "funneled internal os" = cervical incompetence
- Cervical incompetence $\rightarrow$ premature cervical shortening $\rightarrow$ bleeding as cervix is stretched

32) Cyclic progestin therapy

- This patient likely has irregular menses due to anovulation - endometrium is stuck in proliferative phase due to missing ovulation/progesterone
- OCPs are contraindicated because this patient is a smoker >35 years of age (and has HTN)

33) Hysterosalpingography

- Hysterosalpingography rules out tubal occlusion and assesses the uterine cavity
- Multiple sexual partners $\rightarrow$ STIs $\rightarrow$ PID $\rightarrow$ adhesion/obstruction
- She has regular periods indicating that her HPO axis is intact, so we don't need LH, FSH, Progesterone or Estrogen tests

34) Pulmonary hypoplasia

- Oligohydramnios $\rightarrow$ lack of normal alveolar distention by aspirated amniotic fluid $\rightarrow$ pulmonary hypoplasia

35) Type 2 diabetes mellitus

- 'Velvety pigmented skin over the axillae' = Acanthosis nigricans
- Acanthosis nigricans is associated with insulin resistance $\rightarrow$ $\uparrow$ risk of DMII

36) Wound infection

- Erythema and induration indicate wound infection as the cause of this patient's post-op fever

37) Laparoscopy

- Laparoscopic visualization is the only way to definitively diagnose endometriosis

38) Respect the patient's wishes and schedule a follow-up visit in 1 week

- Parental consent is not required for: contraception, STIs, pregnancy, substance abuse, and emergencies/trauma

39) Ultrasonography

- This patient has a suspected twin pregnancy
- IVF (oocyte donation) is a risk factor for twin pregnancy, and this patient has a fundal height > gestational age

40) Decreased estrogen and follicle-stimulating hormone (FSH) concentrations

- This patient has functional hypothalamic amenorrhea
- Excessive training, low-calorie diet, anorexia → ↑ Ghrelin, ↓ Leptin → ↓ GnRH → ↓ LH, ↓ FSH → ↓ Estrogen → amenorrhea, bone loss

41) Antibiotic therapy if delivery has not occurred 18 hours after rupture of membranes

- Indications for Intrapartum GBS prophylaxis:
 - 1) previous infant with invasive GBS disease
 - 2) GBS bacteriuria during any trimester of current pregnancy
 - 3) ⊕ GBS screen culture during current pregnancy
 - 4) Unknown GBS status at the onset of labor plus any of the following:
 - Delivery at <37 weeks gestation
 - Intrapartum fever
 - Rupture of amniotic membranes for ≥18 hours

42) Nonstress test

- Smoking → uteroplacental insufficiency → IUGR
- Nonstress test should be the next step for fetal monitoring; late decelerations indicate uteroplacental insufficiency

43) Vaginal foreign body

- Should be managed by irrigation with warmed fluid (after application of a topical anesthetic in the vaginal introitus) or removal with a swab
- Speculum examination should *not* be performed in prepubertal girls

44) Gestational diabetes

- Previous big baby (macrosomia)

45) Androstenedione to estrone

- Androstenedione is converted to estrone in adipose tissue via aromatase (this patient has a BMI of 34)
- ↑ estrogen → ↑ risk of endometrial cancer

46) Fluorescent treponemal antibody absorption test

- Next step for ⊕ VDRL is FTA-ABS

47) Uteroplacental insufficiency

- HTN in mother → Intrauterine growth restriction (IUGR)

48) Polyhydramnios

- Diabetes → polyhydramnios → fundus larger than gestational age
- This patient has received prenatal care since 7 weeks gestation and has had no abnormalities until 24 weeks gestation - multiple gestations, pelvic tumors, and errors in gestational age would have been picked up before this point in time.

49) Androgen insensitivity syndrome

- X-linked mutations of androgen receptor; Karyotype 46,XY
- Differentiated from Mullerian agenesis (karyotype 46, XX) by lack of axillary & pubic hair

50) *Haemophilus ducreyi*

- Large deep ulcers with gray/yellow exudate, well-demarcated border, soft friable base
- Other painful genital lesion is HSV (genital herpes), which is not an answer choice
- Granuloma inguinale (donovanosis), Syphilis, and Lymphogranuloma venereum have non painful initial lesions